

Wave and Optics

Part-3

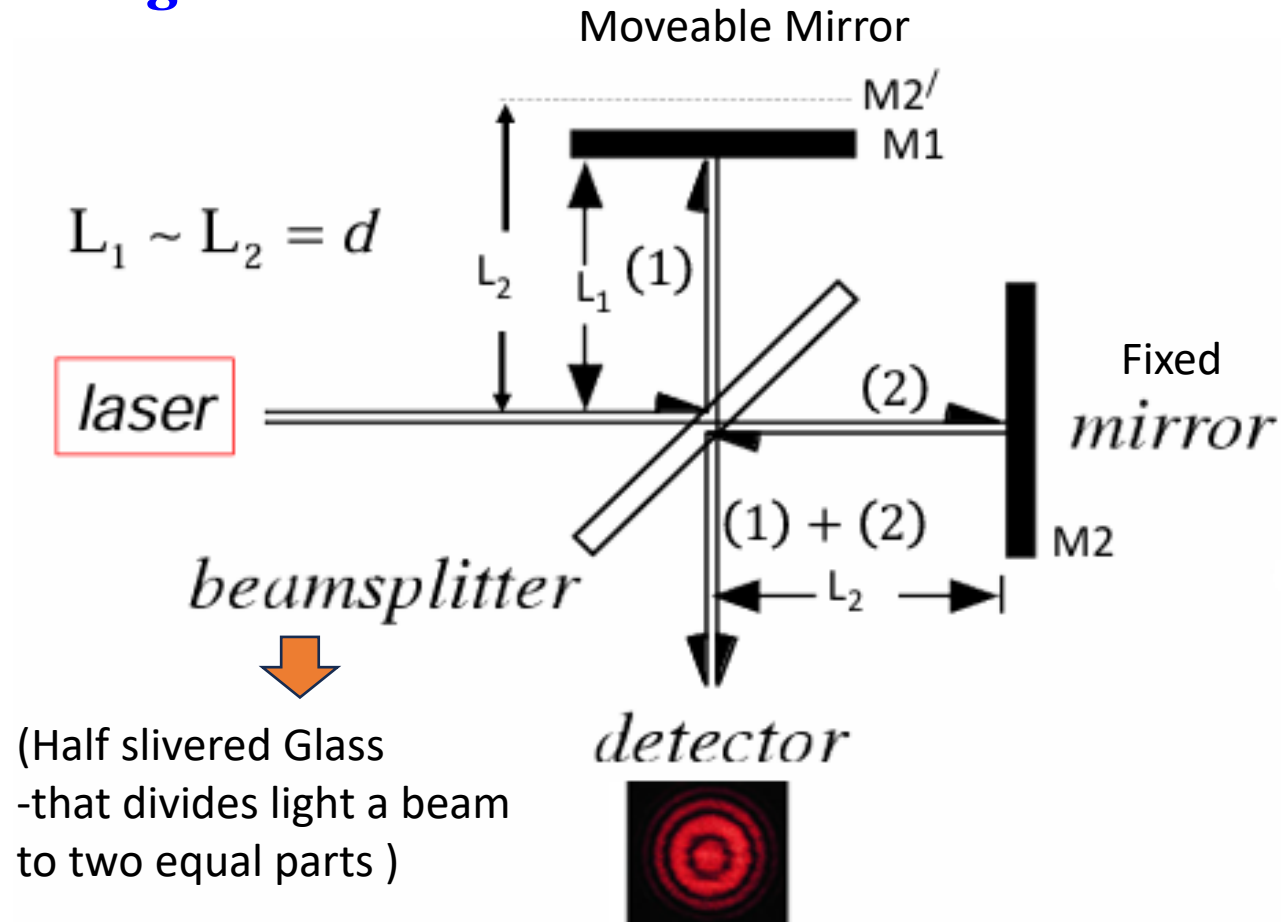


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Michelson interferometer

It is an optical instrument that splits a beam of light in two, sends the parts along perpendicular paths, and then brings them back together.

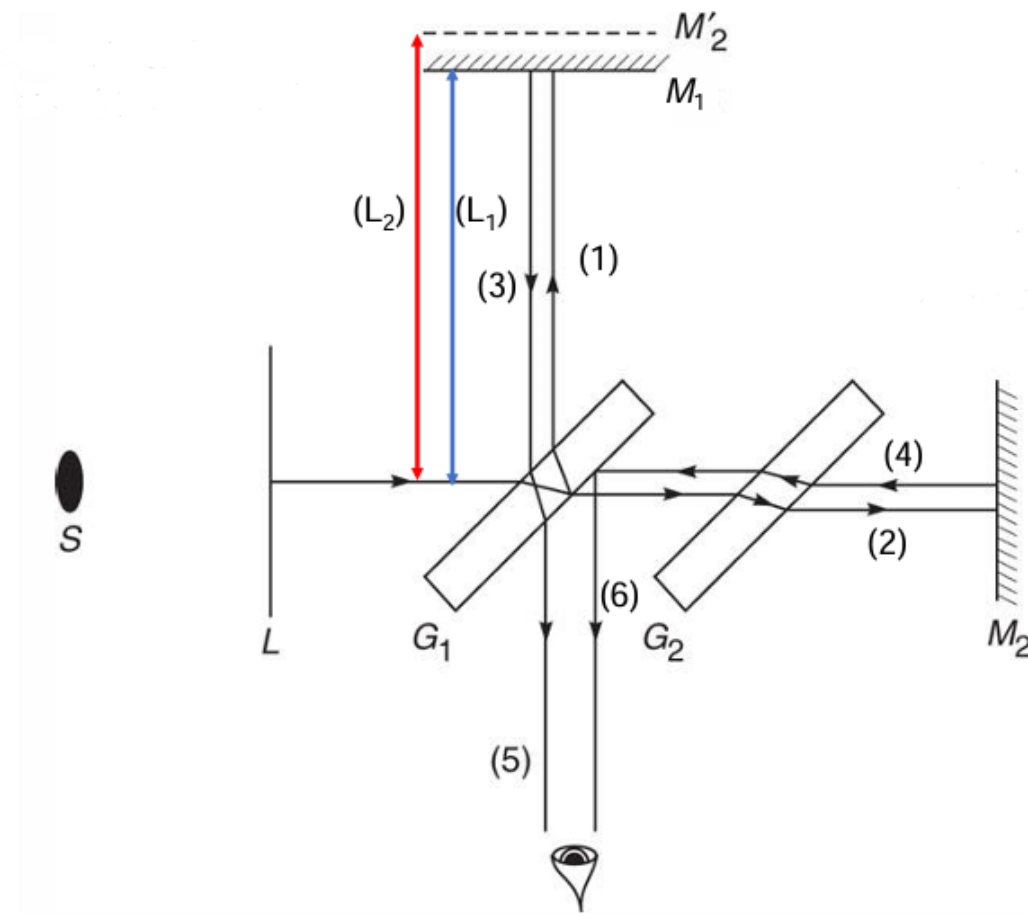


- Light traversing optical paths (1) and (2) interfere at the detector plane/eye
- For the detector, image of M2 will appear at M2' depending on difference in the geometrical length L_2 from L_1

□ When the difference between the two paths is equal to an integral number of light wavelengths, the two beams interfere constructively

□ When the difference is half an integral number of light wavelengths, the two beams interfere destructively

What is the role of the compensating glass plate ?



- Glass plates $G_{1,2}$ are of same material and of same thickness
- Rays (5) and (6) traverse the same path length in glass

HOW ?? Check !!

$$L_1 \sim L_2 = d$$

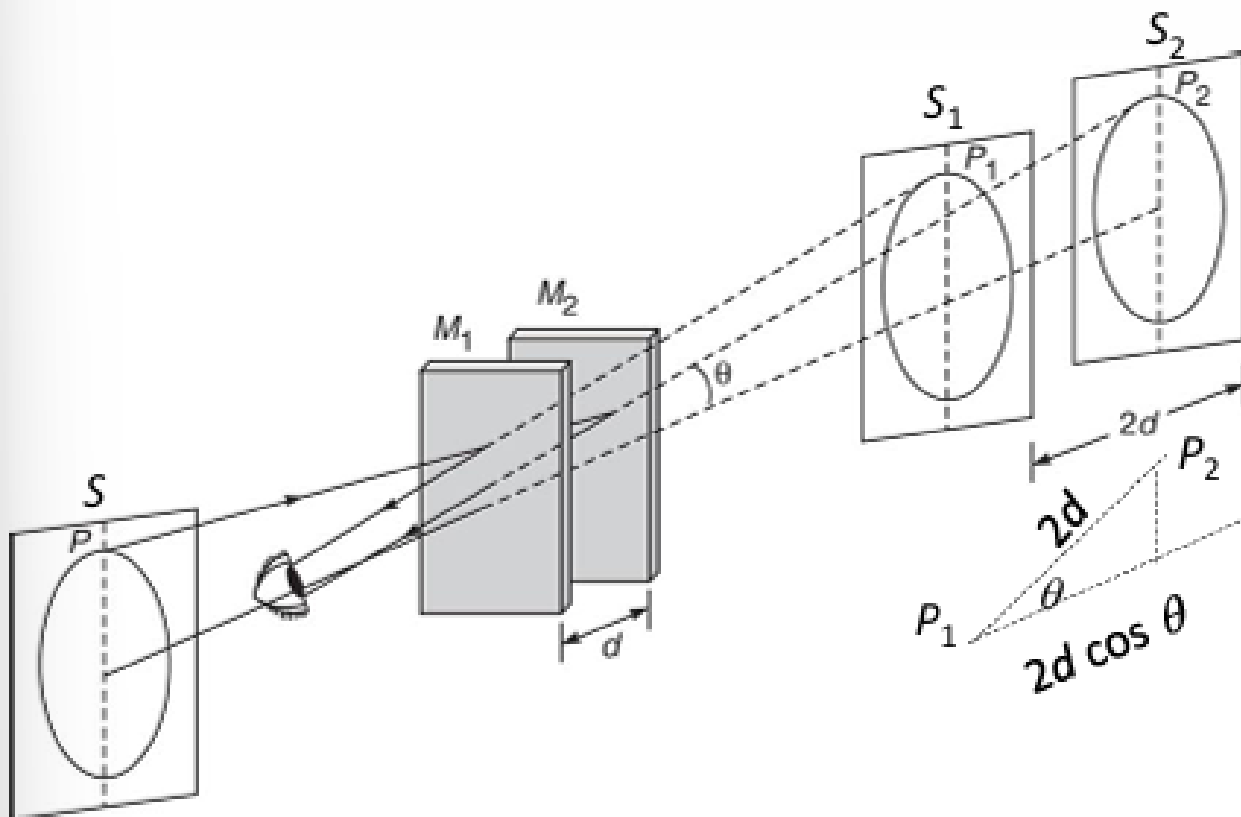
⇒ For maxima

$$2d \cos \theta = m\lambda$$

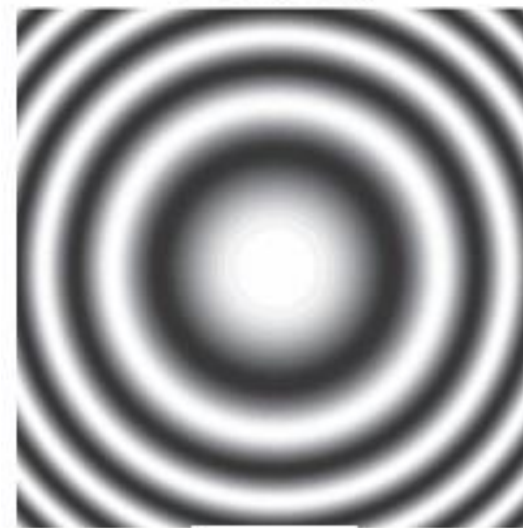
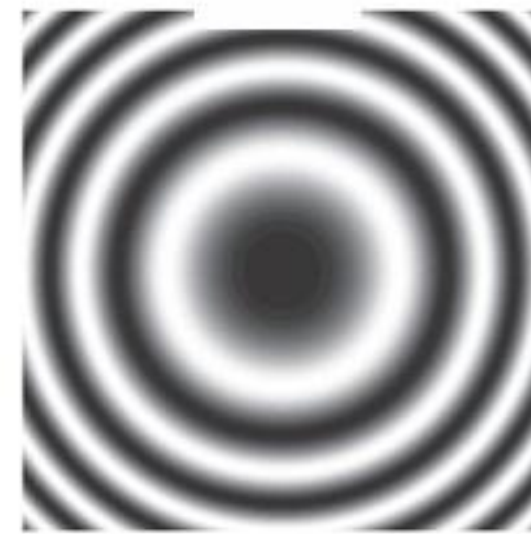
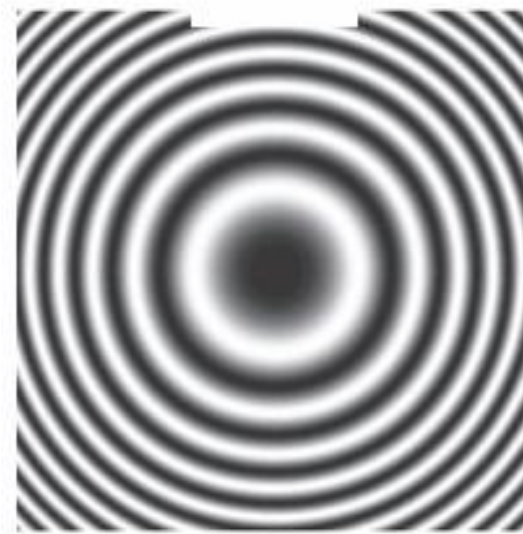
θ is the angle rays make with the axis (normal to the mirror)

The horizontal beam crosses the beam splitter three times, while the vertical beam crosses the beam splitter once.

To equalize the dispersion, a so-called compensating plate identical to the substrate of the beam splitter may be inserted into the path of the vertical beam.



- Examples:
- 1). The Earth's motion through the supposed luminiferous aether that most physicists at the time believed was the medium in which light waves propagated.
 - 2). Gravitational wave detection
 - 3). Fourier transform spectrometer



Computer generated interference patterns from a Michelson interferometer for different d

Example :

For a given $\lambda = 0.6\mu m$, $d = 150\mu m$

Find m for which maxima (circular fringes) appear in a Michelson interferometer

$$(2 \times 150) \cos \theta = m \times 0.6$$

$$\Rightarrow \cos \theta = \frac{0.6}{2 \times 150} \times m$$

$$\Rightarrow \cos \theta = 0.002 \times m$$

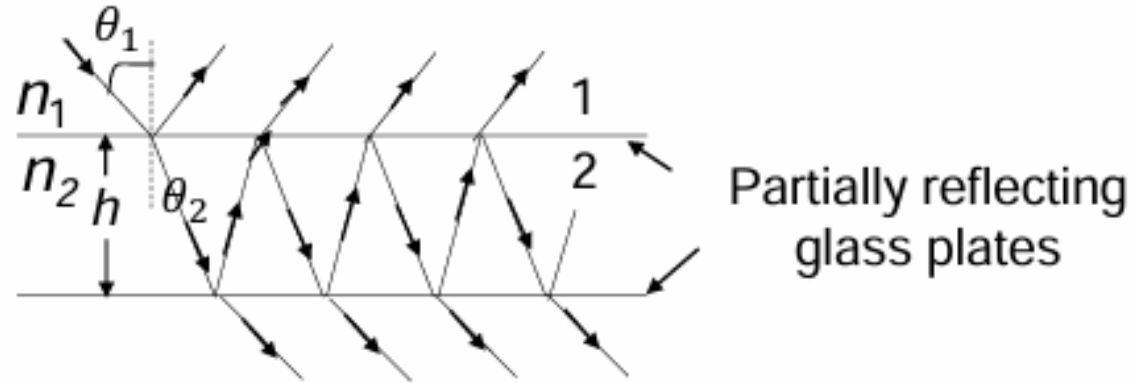
$$\Rightarrow \theta = 0^\circ, m = \frac{1}{0.002} = 500 !$$

$$\Rightarrow \text{For } m = 499, \theta = \cos^{-1}(0.998) = 3.62^\circ$$

$$\Rightarrow \text{For } m = 498, \theta = \cos^{-1}(0.996) = 5.126^\circ$$

Multiple beam interferometry

A third variety of interferometry involves multiple beams derived from same source through divisions of amplitude by multiple reflections



Consider incidence of a plane wave of amplitude A_o from medium 1 of R.I. n_1 at an angle θ_i on a glass plate of thickness h and of R.I. n_2

If r_1 and t_1 represent reflection and transmission coefficients of the plate as incident light from medium 1 enters medium 2, then amplitudes of successive reflected waves will be

$A_o r_1$, $A_o t_1 r_2 t_2 e^{i\delta}$, $A_o t_1 r_2 r_2 r_2 t_2 e^{2i\delta}$, ... (Calculate these -Will be part of your assignment)

where

$$\delta = \frac{2\pi}{\lambda_o} \times \Delta = \frac{2\pi}{\lambda_o} \times (2hn_2 \cos \theta_2) = \frac{4\pi hn_2 \cos \theta_2}{\lambda_o}$$

⇒ Resultant amplitude of the reflected waves from the interface between the medium 1 and medium 2 :

$$A_r = A_o \left(r_1 + t_1 t_2 r_2 e^{i\delta} \left[1 + r_2^2 e^{i\delta} + r_2^4 e^{2i\delta} + \dots \right] \right) = A_o \left(r_1 + \frac{t_1 t_2 r_2 e^{i\delta}}{1 - r_2^2 e^{i\delta}} \right)$$

$$A_r / A_o = r_1 \left(1 - \frac{(1 - R) e^{i\delta}}{1 - R e^{i\delta}} \right)$$

If the reflectors are lossless, the reflectivity and the transmittivity at each interface are given by

$$R = r_1^2 = r_2^2$$

$$\tau = t_1 t_2 = 1 - R$$

where we have used the fact that $r_2 = -r_1$

⇒ Net reflectivity $\mathcal{R}$:

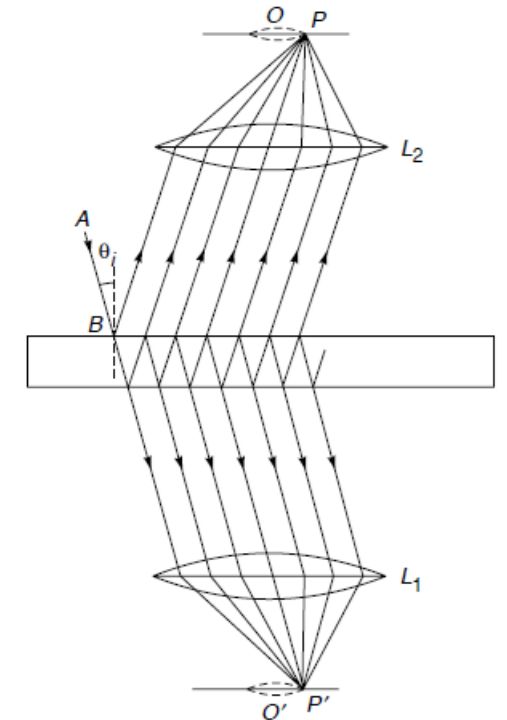
$$\left| \frac{A_r}{A_o} \right|^2 = R \left| \frac{1 - e^{i\delta}}{1 - R e^{i\delta}} \right|^2$$

$\mathcal{R}$ can be shown to be :

$$\mathcal{R} = \frac{F \sin^2 \frac{\delta}{2}}{1 + F \sin^2 \frac{\delta}{2}}$$

where

$$F = \frac{4R}{(1-R)^2} ; F \text{ is called coefficient of finesse}$$



For $R \ll 1$, F is small $\Rightarrow \mathcal{R} \propto \sin^2 \frac{\delta}{2}$

Similarly, amplitude of successive transmitted waves (assuming zero phase for the 1st transmitted wave):

$$A_o t_1 t_2, A_o t_1 t_2 r_2^2 e^{i\delta}, \dots$$

It can be shown that transmittivity T of the film/glass plate will be given by

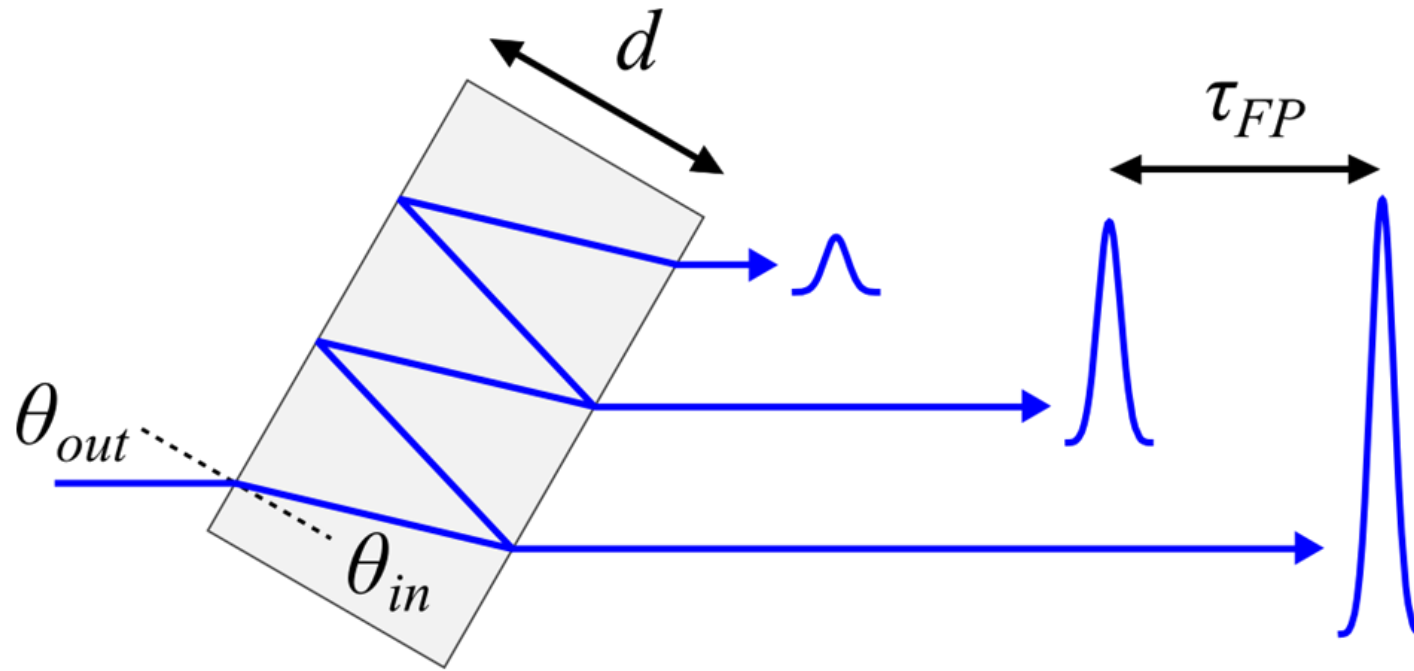
$$T = \left| \frac{A_t}{A_o} \right|^2 = \frac{1}{1 + F \sin^2 \frac{\delta}{2}} \quad \text{(Will be part of your assignment)}$$

Also for, $\delta = 2m\pi$, $m = 1, 2, 3, \dots$

$$T = \frac{1}{1 + F \sin^2 m\pi} = 1$$

Example:
THE FABRY-PEROT ETALON

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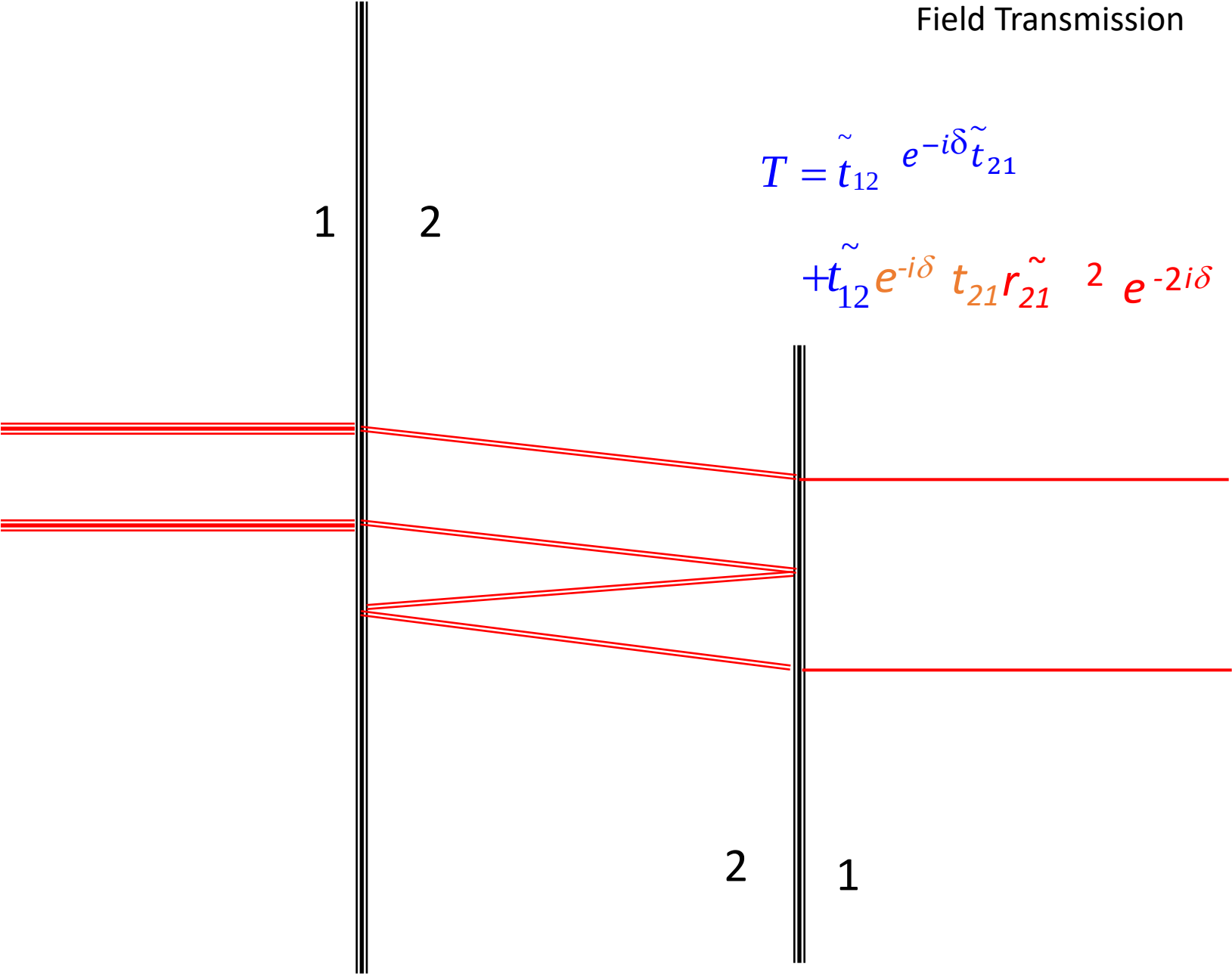


Fabry-Perot

Field Transmission

$$T = \tilde{t}_{12} e^{-i\delta} \tilde{t}_{21}$$

$$+ \tilde{t}_{12} e^{-i\delta} t_{21} \tilde{r}_{21}^2 e^{-2i\delta}$$



Thanks